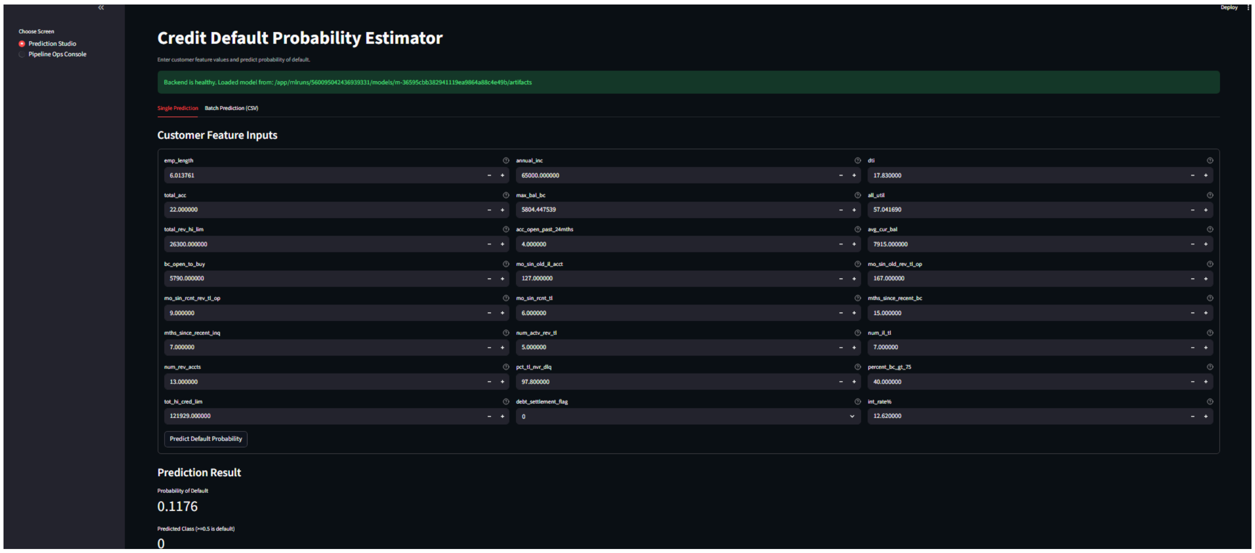
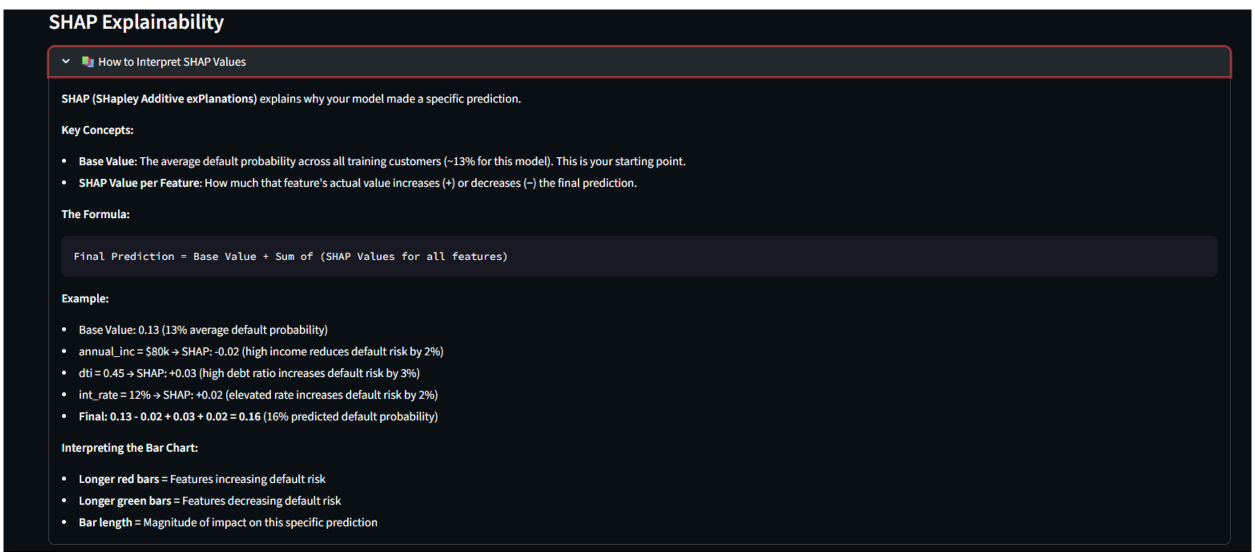
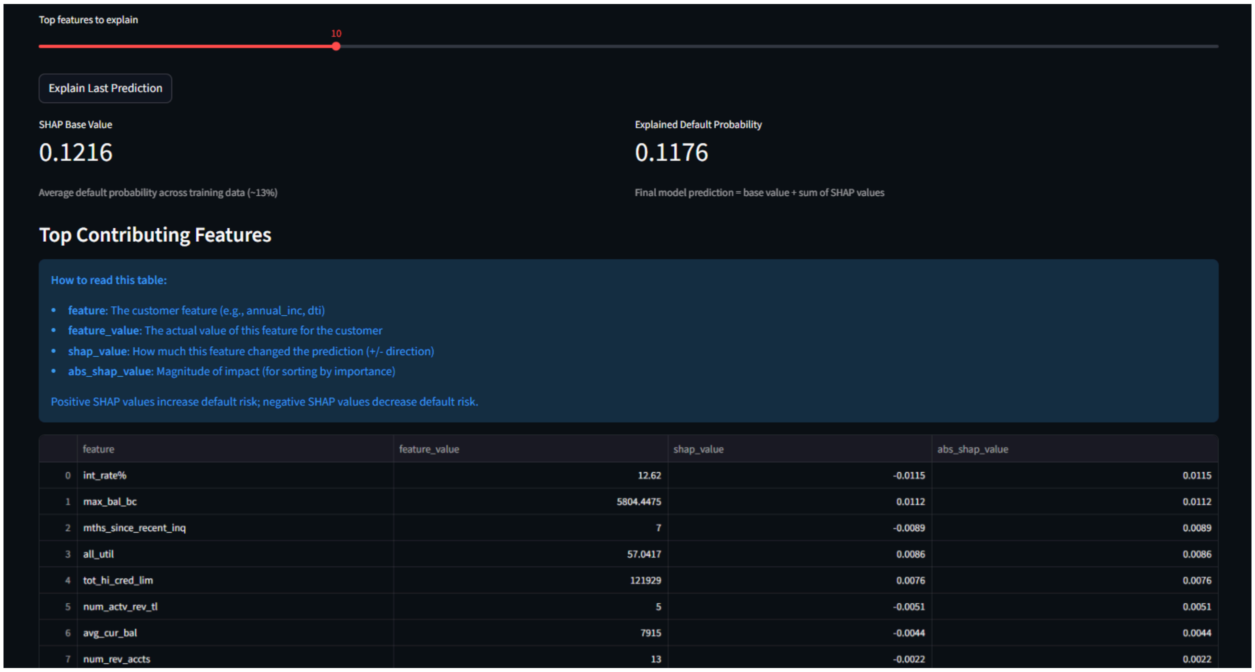
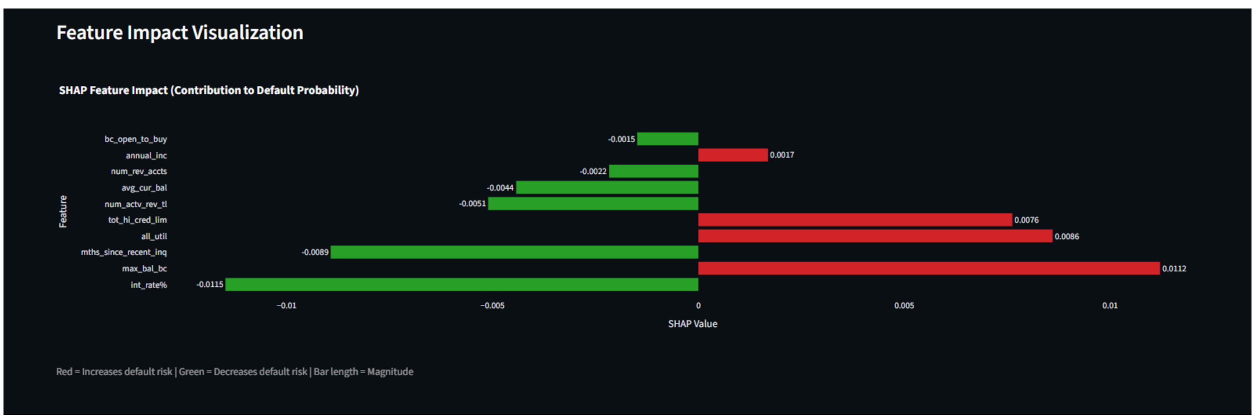
User Guide: Credit Risk Platform Interface







	feature	feature_value	shap_value	abs_shap_value
0	int_rate%	12.62	-0.0115	0.0115
1	max_bal_bc	5804.4475	0.0112	0.0112
2	mths_since_recent_inq	7	-0.0089	0.0089
3	all_util	57.0417	0.0086	0.0086
4	tot_hi_cred_lim	121929	0.0076	0.0076
5	num_actv_rev_tl	5	-0.0051	0.0051
6	avg_cur_bal	7915	-0.0044	0.0044
7	num_rev_accts	13	-0.0022	0.0022



1. Overview

The Credit Risk Platform provides a web-based interface designed to support two primary workflows:

1. **Credit Risk Prediction and Explainability**
   - Perform single and batch predictions
   - Analyze model outputs and feature contributions

2. **Pipeline Operations and Monitoring**
    - Monitor system health
    - Trigger pipeline executions
    - Track experiments and performance metrics

The interface consists of two main modules:

- **Prediction Studio**
- **Pipeline Operations Console**

Prediction Studio is used for model inference tasks, while the Pipeline Operations Console provides visibility and control over the MLOps system.

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2. Access and Prerequisites

Before accessing the interface, ensure that the following services are running:

- Backend inference API
- Frontend application
- Prometheus (metrics collection)
- Grafana (visualization)
- MLflow (experiment tracking)
- Airflow (optional, for orchestration features)

The frontend can be accessed through the configured Streamlit URL.
If the backend service is unavailable, prediction functionality will be disabled and an error message will be displayed.

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3. Navigation

The application uses a sidebar-based navigation system.

Users can switch between:

- **Prediction Studio**
- **Pipeline Operations Console**

Switching between screens does not require restarting services.

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4. Prediction Studio

4.1 Purpose

Prediction Studio enables:

- Single customer risk prediction
- Batch prediction using CSV files
- Model explainability using SHAP

4.2 Initial Health Check

Upon loading, the interface performs a backend health check and retrieves:

- Required feature list
- Feature descriptions
- Default values
- Binary feature indicators
- Model reference

If the health check fails, prediction actions are disabled.

4.3 Single Prediction

A dynamic input form is generated based on required model features.

Steps:

1. Enter values for all input features
2. Select 0 or 1 for binary features
3. Enter valid numeric values for continuous features
4. Click **Predict Default Probability**

Output:

- Probability of Default (PD)
- Predicted class (threshold = 0.5)

- Model metadata

Important Notes:

- A warning is shown if many fields are left as zero
- This prevents unrealistic input scenarios

4.4 Explainability (SHAP)

After generating a prediction, users can view explainability insights.

Steps:

1. Select number of top features
2. Click **Explain Last Prediction**

Output:

- SHAP base value
- Final predicted probability
- Feature contribution table
- Visualization of feature impacts

Interpretation:

- Positive SHAP → increases default risk
- Negative SHAP → decreases default risk
- Larger magnitude → stronger influence

4.5 Batch Prediction

Supports prediction for multiple records using CSV input.

Steps:

1. Upload CSV file
2. Validate column alignment
3. Preview input data

4. Run batch prediction

Output:

- Default probability per row
- Predicted class
- Downloadable results

Validation Behavior:

- Extra columns are ignored
- Missing required columns trigger errors

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5. Pipeline Operations Console

5.1 Purpose

The Pipeline Operations Console provides centralized monitoring and control for:

- System health
- Pipeline execution
- Experiment tracking
- Performance monitoring

5.2 Auto-Refresh

A configurable refresh interval allows automatic updates of system status.
Setting it to zero enables manual refresh only.

5.3 Platform Health Monitoring

Displays status (UP/DOWN) for:

- API
- Prometheus
- Grafana
- MLflow
- Airflow

Each service includes endpoint details or error summaries.

5.4 Pipeline Management

Provides controls to execute pipeline operations:

- Run DVC pipeline (`dvc repro`)
- Trigger MLflow experiments
- Check pipeline status

Outputs:

- Execution logs
- Success/failure status

5.5 Error and Run Tracking

Tracks pipeline and experiment execution:

- Airflow DAG run states
- MLflow experiment status

Includes:

- Run history tables
- Failure detection

5.6 Speed and Throughput Monitoring

Displays performance metrics:

- Requests per second
- P95 latency
- Error rate (5xx)
- Average drift score

If metrics are unavailable, a warning is shown.

5.7 Tool Navigation

Provides direct links to:

- Airflow UI
- MLflow UI
- Prometheus UI
- Grafana UI

5.8 Experiment Metadata

Allows inspection of the latest experiment metadata, including run identifiers and model context.

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6. Typical User Workflows

6.1 Business Analyst Workflow

1. Open Prediction Studio
2. Enter customer details
3. Generate prediction
4. Analyze SHAP explanation
5. Export results

6.2 ML Engineer Workflow

1. Open Pipeline Ops Console
2. Verify system health
3. Trigger pipeline or experiment
4. Monitor execution status
5. Analyze performance metrics

6.3 MLOps Monitoring Workflow

1. Check service availability
2. Monitor latency and throughput
3. Analyze drift metrics
4. Use Grafana dashboards for insights

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7. Input and Data Guidelines

- Provide realistic input values
- Avoid excessive zero entries
- Ensure CSV schema matches required features
- Maintain clean numeric data

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8. Troubleshooting

8.1 Backend Connection Issues

- Ensure API is running
- Check logs for errors

8.2 MLflow Access Issues

- Service may require authentication
- Restricted endpoints are expected in secure setups

8.3 CSV Validation Errors

- Verify required columns
- Correct column naming

8.4 Metrics Not Available

- Check Prometheus configuration
- Ensure API metrics endpoint is reachable

8.5 Performance Issues

- Reduce active services
- Check system resource usage
